AgentWorkShop: A Governed Supervisory Execution Layer for LLM-Agent Plant Writes

Anonymous Authors

***Abstract*—Large language model (LLM) agents increasingly propose control changes on industrial plants, yet the safeguards surrounding each proposal—write authority, active limits, human approval, device acceptance—are rebuilt per deployment inside application code, so a task reported complete can coexist with a missed process objective. AgentWorkShop, an open-source governed supervisory execution layer, carries every agent-originated write along one path: admission against declared, versioned limits; approval where required; actuation over semantic write-control nodes; graded acknowledgment—readback-verified, transport-acknowledged, or unverified—by what an independent post-write read observes, not the transport receipt alone; and verdict recording with compensation—each stage a separately inspectable evidence state attributed to its originating actor. Team identities reach data-acquisition and write-control nodes only through explicit member–node capabilities, prompt content is never admissible as authority, and a fail-safe-manual line master switch intercepts approval-free recipe dispatch. Four process scenarios—injection molding, wastewater treatment, annealing, and biaxially oriented polyethylene terephthalate (BOPET) film—and a cast-film reference line exercise the layer across protocol-interception drills, agent-team missions, and a deployment-scale regression campaign on one multi-line instance. The evaluation is simulation-tier (simulator-backed, not physical plants); no optimization-benefit or safety-assurance claim is made.**

***Index Terms*—Agent teams, human-in-the-loop, industrial informatics, LLM agents, supervisory control, write governance.**

I. Introduction

LARGE language model (LLM) agents are moving from advising industrial operations to acting on them: recent systems generate control updates from multi-modal input [1], pair each factory module with an LLM agent that dispatches skills over a tool protocol [2], allocate work among cooperating agents [3], [4], and act as real-time controllers of a robot line [5]. What these demonstrations leave implicit is everything that surrounds the proposal: who may change which parameter, which limits are active for the current product and recipe, whether a human approved, whether the device accepted the command, and whether the process improved.

Because these concerns live in application code, they are rebuilt per process, protocol, and agent framework—duplicated drivers, ad-hoc permission checks, logs untied to decisions. The four scenario plants (simulated in this paper)—injection molding, wastewater treatment, annealing, and BOPET (biaxially oriented polyethylene terephthalate) film—differ, but all need an auditable bridge from a team-level task to heterogeneous acquisition and control nodes.

AGENTWORKSHOP—an open-source governed supervisory execution layer (Section III)—addresses this gap through member–node bindings (a member being an engine-backed channel identity) and shared line, product, recipe, and run context. The response is organizational rather than algorithmic: each intervention leaves a separate evidence record per governed stage, from admission through compensation (Section IV). Nodes expose engineering quantities rather than register addresses, and one governed member–node binding path serves every client.

AGENTWORKSHOP makes four contributions:

1) a **capability-based industrial execution architecture** connecting agent-team identity, semantic data-acquisition (DAQ) and write-control (DCW, the platform's node-class tag) nodes, heterogeneous protocol drivers, and line–product–recipe–run context, with task membership distinct from physical-write authority and a default-manual line master switch;
2) a **reusable integration boundary** of self-describing node and driver contracts across six protocol families (five instantiated as lines, four exercised for governed writes), versioned production objects, plugin extension points, and a typed software development kit (SDK); an onboarding contract projects scenario blueprints into it while process models and policies stay external;
3) an **evidence-preserving governed intervention workflow** separating admission, optional approval, actuation, graded write acknowledgment with post-write verification, verdict, compensation, and backstop—the grading rule stated falsifiably; and
4) a **layered evaluation** of one governed path across protocol closure with mechanism ablation, agent-team missions, a deterministic controller loop, concurrent process missions, and a deployment-scale regression campaign on a multi-line instance (Section V).

Scope, revisited in Section VI: real protocol stacks execute against author-implemented simulators; fast regulation, certified protection, physical-plant transfer, and optimizer ranking are outside the claims. The governed path is proposer-agnostic—prompt content is never authority and admission re-resolves per request—so the scripted proposers of Section V exercise the same contract an engine-issued proposal would face. The remainder of this paper proceeds from the nearest systems (Section II) and the platform architecture (Section III) through the governed write path (Section IV) and its layered evaluation (Section V) to validity, safety positioning, deployment boundaries, and conclusions (Sections VI and VII).

TABLE I
CLOSEST LLM-BASED INDUSTRIAL CONTROL SYSTEMS VERSUS AGENTWORKSHOP (VENUE, LLM ROLE, WRITE PATH, PRODUCTION-SCOPED AUTHORITY, WRITE EVIDENCE).

System (venue)	Role of the LLM	Plant write path	Authority & production context	Write evidence & verification
MADE, ICCAD 2026 [1]	multi-modal control-update generation	generated commands + safety layer	HITL verification of outputs; task-level checks	task success over scripted evaluations
Köhle et al., ETFA 2026 [2]	per-module agents, MCP skill dispatch	skill invocation via OPC UA/MQTT	module-scoped skills	task-resolution rate (6-module plant)
Lim & Kovalenko, CASE 2025 [3]	capability exploration across agents	architecture-level	resource-matching scope	architecture evaluation
Zhao et al., AEI 2026 [4]	shop-floor task dispatch	workshop experiments	coordination layer	dispatch/execution performance
Waseem et al., Mach. Learn. Eng. 2026 [5]	real-time controller in the loop	direct loop actuation (robot line)	prompt-level constraints	line KPIs under LLM control
AGENTWORKSHOP	**replaceable strategy slot**	**governed write path: admission, HITL, dispatch, readback**	**member–node bindings; line master switch; product/recipe-scoped limits**	**graded acknowledgment; harness-checked three-way consistency; attributed journal; compensation; backstop**

II. RELATED WORK

A. LLM Agents in Industrial Control: From Generation to Governance

Agent-based manufacturing and supervision have long linked organizational roles to plant capabilities [6], [7], and LLM-oriented industrial work now spans PLC generation and verification [8] and direct plant-facing action—context-aware control [9], digital-twin fault handling [10], and natural-language machine control over OPC UA [11]—without production-scoped admission, per-request re-checking, or graded write evidence. The closest systems (Table I, as reported in their publications) target the control action itself—multi-modal control-update generation behind a safety layer [1], module-scoped skill dispatch over a six-module plant [2], and pretrained LLMs *inside* the control loop as real-time controllers [5], a layer below AGENTWORKSHOP so the roles are complementary—while agent-allocation work [3], [4] distributes tasks, not governed writes. A recent review frames agentic AI's industrial value as safety-aware human-in-the-loop (HITL) decision support rather than autonomous control authority [12]—the mechanism set here makes that concrete.

B. Frameworks, Standards, Run-Time Guarantees, and Automation Practice

The Model Context Protocol (MCP) standardizes multi-agent tool interfaces [13]; its tools carry advisory annotations but no production context—line, product, or recipe—and no record separating admission from execution. ISA-88 and ISA-95 model recipes and operations [14], [15], and AGENTWORKSHOP's versioned recipes instantiate a reduced subset of that stack; the governed life cycle transposes run-to-run (R2R) control disciplines [16] onto a supervisory write path. A separate line enforces run-time guarantees on LLM agents—guard agents [17], schema-level behavioral constraints (AgentSpec [18]), and capability-based information-flow control (CaMeL [19]), the closest software-agent analogue of the rule that prompt content is never authority; they gate software tool calls and data flow, not plant writes; their policies could compile onto this substrate.

Permissions, range checks, approvals, and audit trails long existed in distributed-control-system (DCS) and SCADA practice—versioned batch recipes and electronic batch journals under 21 CFR Part 11 and GAMP 5 [20], [21], second-channel monitoring and advisory operations in NAMUR open architecture [22], and setpoint adaptation in run-to-run control—while IEC 62443 codifies their security counterpart [23] and formal shielding, control-barrier functions, and simplex run-time assurance offer stronger properties [24]–[26]. Graded command confirmation itself has telecontrol ancestry: DNP3's select-before-operate exchange [27] and IEC 60870-5's command-acknowledgment cycle [28] confirm a command at the protocol and application layer; AGENTWORKSHOP grades above them, against production context and actor attribution. Against that practice the difference is the proposer: a machine-generated agent identity rather than a presupposed human operator, so admission binds to member authority, prompt content stays inert, and intervals are re-evaluated per request against live product and recipe state; the evidence states—graded acknowledgment, verdict, compensation—survive strategy replacement; the three-way check is a supervised special case of classical data reconciliation [29]. AGENTWORKSHOP implements source-dependent checks and a rule-based backstop (Section IV-D), no formal protection guarantee; human oversight likewise requires more than an approval dialog [30].

III. SYSTEM ARCHITECTURE

AGENTWORKSHOP occupies the replaceable strategy slot that Section II's comparison leaves open: it links team coordination to plant interaction through industrial nodes and production context (Fig. 1). Responsibilities group into five planes—logical groupings, not deployable units: a *team* plane owns member identity, task state, and episodic memory; an *integration* plane owns protocol acceptance, readback, freshness, and transport errors, presenting devices as semantic nodes feeding both the agent and a live plant-mirror view; a *governance* plane owns the governed life cycle—admission through compensation and rollback; an *operations* plane attaches line, product, recipe, and run context while PLC protection stays independent; and an *extensibility* plane admits plugins, drivers, agent tools, and SDK clients through sanctioned registration services. Scenario knowledge sits in blueprints, signal maps, predicates, and driver configuration—not in scheduler, approval, or persistence code; members are

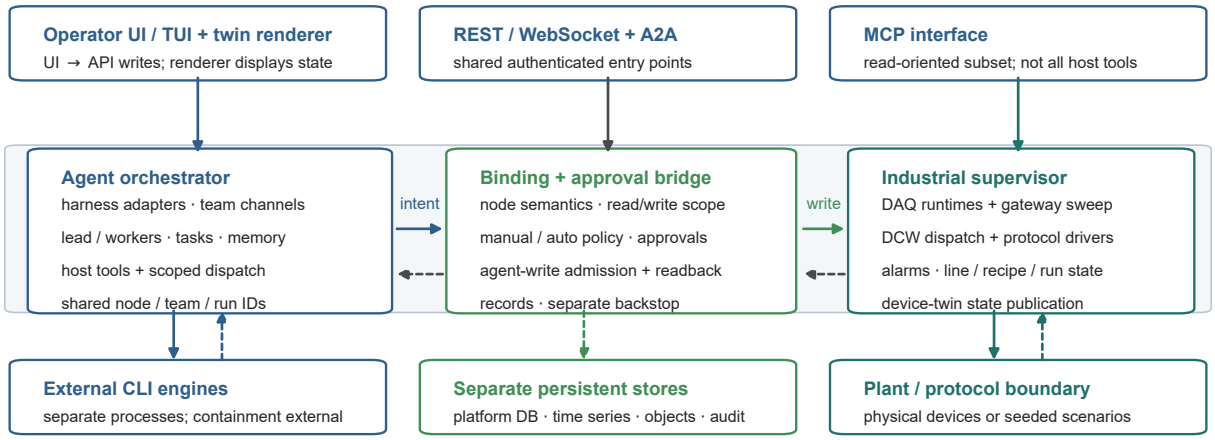


Fig. 1. System architecture and execution boundaries: platform services around shared stores, with agent engines, data stores, protocol endpoints, and the plant kept separate. Shared identifiers provide attribution but do not make storage and actuation transactional. Solid arrows: request/control; dashed: observation/record; PLC safety and hard real-time control remain external.

interchangeable across engine adapters, and team topology, bindings, and limits are data rather than code—in every plane, a scenario supplies data and strategy (Fig. 1).

As a deployable project, AGENTWORKSHOP is open source under the Apache-2.0 license,[1] and ships the planes above as one service—team and channel runtime, multi-protocol acquisition, governed write control, versioned recipes and run ledgers, an adaptive-modeling training loop with candidate-to-production promotion, and plugin bridges to external knowledge-base and diagnosis services—with 58 industrial agent tools and the typed SDK declared over the same node and driver contracts, and the one-command benchmark pipeline of Section V-B in the distribution.

A. Supervisory Core and Nodes

The Node.js core hosts channel routing, industrial services, and protocol adapters. A channel contains a lead, workers, tasks, messages, and retrievable memory, deployed through WebSocket, MCP, agent-to-agent (A2A), REST, CLI, or TUI entry points; channel membership alone confers no write authority.

DAQ nodes convert device readings to engineering quantities across four modalities—scalars, vector profiles, image frames carrying content fingerprints, and mirrored manufacturing-execution-system (MES) records—so plant data-shape heterogeneity does not reach the team loop. MES endpoints are reached in two modes—*mirror*, pulling MES records through the same acquisition pipeline, and *direct*, declaring read and write mappings against the MES interface. DCW nodes translate engineering setpoints into driver commands and, where supported, readback; per-node runtimes manage timing, and gateway services supply connection state, recipe context, and write bookkeeping. The acquisition surface declares six protocol families, five with drivers today—Modbus TCP, Modbus RTU-over-TCP, OPC UA, MQTT, HTTP/REST, and an S7 stub that fails loudly until its driver lands; the write surface adds one driver per writable family plus MES REST (the evaluation's mock driver lies outside these counts), and the evaluation instantiates five families as lines (Section V-C). The deployment reaches author-implemented dynamic simulators through the same node and driver contracts a plant would use; they are deterministic per seed, not validated digital twins. The first reuse boundary is declarative—signals, units, scaling, precision, and driver configuration—and the team loop is unchanged.

Driver contracts are deliberately narrow: a driver declares how a semantic quantity maps to its transport payload, whether the endpoint is writable, and whether a readback is available, while the controller resolves member authority and active limits. Errors attach to the request, register addresses and byte order appear only here, and a new fieldbus family leaves the loop untouched. Undeclared drivers fail loudly rather than being silently served by a mock. Acquisition is supervisory rather than hard real-time: each DAQ node runs its own cadence under a rotating sweep; the verdict stage queries the timestamped, quality-stamped history, not a blocking read.

B. Bindings, Context, and Reuse Boundaries

A member–node binding grants access to a sanctioned tool on a specific node and selects automatic or approval-mediated execution. Line permissions (none, read-only, operate) authorize binding creation and mutation; runtime invocation is authorized by the persisted capability, not channel membership. Intervals are re-checked every request instead of cached, so revoking a binding or switching the recipe takes effect on the next request; the same path serves a console, a harness, and an agent engine—authority does not depend on the client, only the evidence record differs.

Production objects are versioned rather than overwritten: a recipe is dispatched against a known-good version, reverted non-destructively, and compared with the values actually applied, and an intervention is read against the definition that was active when it was admitted. Plugins register hooks, routes, agent tools, drivers, and panels only through sanctioned services, so an added driver inherits admission, readback, and evidence capture. Scenario reuse is bounded by an onboarding contract: the engineer declares a blueprint (devices, signals,

[1]https://github.com/kingdol666/AgentWorkShop; deterministic plant simulators: https://github.com/kingdol666/plc-node-simulator.

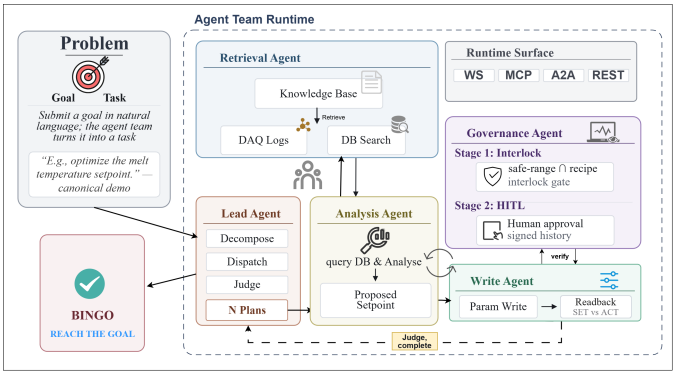


Fig. 2. Implemented agent-team runtime. A natural-language goal enters as a task; the lead agent decomposes and dispatches plans; retrieval and analysis produce the proposed setpoint (the LLM engine occupying a replaceable strategy slot). The governance stages are deterministic platform logic: Gate 1 applies the Eq. (1) interlock, Gate 2 the approval wait; the write agent reads back the value actually present (SET versus observed ACT); the judge records keep, rollback, or uncertain; every step is journaled with actor attribution. "Governance Agent" denotes deterministic platform stages, not a model-bearing agent.

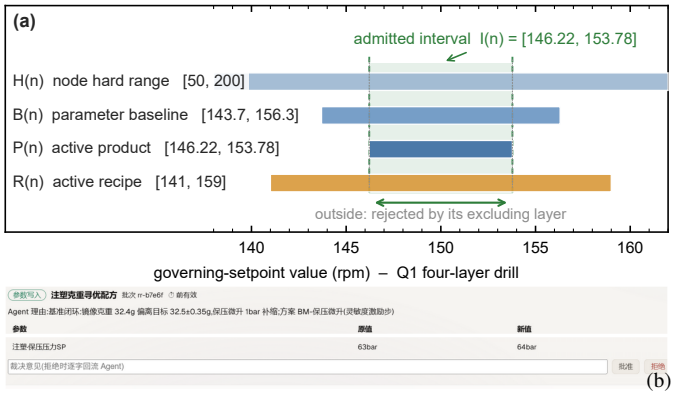


Fig. 3. The admission boundary and its human-approval gate. (a) The Eq. (1) interval geometry on the cast-film reference line: four declared, versioned bounds narrow to the admitted interval; a proposal outside it is rejected by its excluding layer (the Q1 four-layer drill). (b) The structured approval card on a manual binding (live system): machine-checkable fields separated from the proposer's display-only reasoning; the comment returns to the member on refusal (Section IV-E).

units, roles, setpoints); a differential "ensure" pass creates missing devices, repairs drift, and projects the blueprint into semantic nodes and a recipe; a channel is built from the team template, and a mission supplies objective, guards, response windows, and policy. A BOPET worker proposes casting-speed, fast-roll, or rail-width changes through bound DCW nodes; the other three scenarios reuse the same pattern (Section V-F). The stores share identifiers, not a transaction domain: a device may accept a write even if bookkeeping fails.

C. Governed Optimization Task Life Cycle

A governed optimization task is a life cycle rather than a single tool call, and AGENTWORKSHOP exposes each transition separately. The task enters a channel carrying goal, line, product, recipe, run, and task identifiers; a lead member claims it and hands it to a worker holding member–node bindings on the relevant nodes, whose fixed job then repeats: read the controlled quantity over a bounded window, propose a setpoint inside the resolved interval, invoke the governed write, and record a verdict (Section V-B instruments it).

Each iteration leaves more than a written value: an optimization record, the acknowledgment grade, an attributed journal entry, predicate attainment, and a workflow terminal state (per-intervention detail in Section IV). Two write lifecycles coexist: direct node writes (agent `dcw_control` or admin REST) open these records, while recipe-chain operations (batch dispatch) are gated by the run ledger and open none (Section IV). This bounds what "automatic optimization" means here: a controller, a scripted policy, a conventional optimizer, or an agent engine may occupy the strategy slot, while authority, evidence, and closure stay on the platform.

IV. GOVERNED EXECUTION MECHANISMS

A supervisory intervention must preserve distinctions that an agent tool call usually collapses—permission to propose, admission of a value, driver acceptance, device confirmation, process response, a caller's verdict, and compensation—and AGENTWORKSHOP exposes these as separate evidence states. Three design commitments generate that separation: model output is an untrusted proposal, never authority; evidence is graded by what is observed, not claimed; and failure states surface for inspection rather than being absorbed into a success flag. Fig. 2 shows the implemented agent-team runtime carrying these steps. A governed intervention is one transaction in the audit-trail sense: every step leaves its own evidence record, while execution and recovery are deliberately not atomic with device state (Section VI). Six invariants organize the path—binding-resolved authority (never channel membership or prompt content), Eq. (1) admission, fail-closed approval waits on manual bindings, graded rather than success-flag acknowledgment, checkable three-way consistency, and separately admitted compensation—each specified below; every agent-originated path to the plant passes through the same governed write.

A. Task, Observation, and Admission

The strategy receives a task and goal in production context and decides what to propose and when to stop; the platform governs semantic node parameters—pressure, dosing rate, line speed, rail width—in engineering units.

For an ordinary agent setpoint request on node n, the admission interval is

$$\mathcal{I}_n(t) = \mathcal{H}_n \cap \mathcal{B}_n \cap \mathcal{P}_n(t) \cap \mathcal{R}_n(t), \tag{1}$$

where $\mathcal{H}_n$ is the mandatory hard engineering range (finite and non-empty at node creation) and the remaining terms the baseline, active-product, and active-recipe limits; a missing optional limit contributes no restriction, and t is the admission instant, re-resolved at execution. An emptying configuration rejects every write with the excluding layer named, surfacing the misconfiguration at the first write. The interval form is deliberate: every bound is a declared, versioned object an audit can replay against active product and recipe; rejection is decidable and layer-attributable (Fig. 3a)—decidability, not

a safety certificate. It bounds one parameter at a time—joint windows live in scenario guards—and bounds magnitude, not intent: availability, binding state, and approval policy are checked separately, and in-window but adverse sequences surface at verdict and compensation (Section V-F reports two).

B. Semantic Write, Pacing, and Source-Specific Routes

A semantic card exposes quantity, unit, precision, range, value, freshness, and recipe context; the controller resolves binding and limits, the driver serializes the command, and the acknowledgment grade closes the command path, never the process-response path (Section IV-C). Three pacing windows pace governed loops, never the plant: online writes *on a node*—agent- or human-origin—sit at least 60 s apart with a hold window after success (HTTP 429 inside it); a per-recipe dispatch interval anchored at the *approval* instant is restart-persistent, so alternating identities cannot reset it; and a 300 s per-line trial cadence damps same-line trials from any origin; recipe-dispatch and rollback writes are exempt. `AW_BENCH_MODE=1` admits harness-declared arms that relax the soft window or readback—never the hard range.

Four governed routes—agent tools, human-origin writes, recipe dispatch, rollback compensation—reach one admission and journaling entry point, the hard range included; recipe dispatch adds run context—the line-scoped execution window, the reduced ISA-88 batch/lot notion—validated at save time. Hold heartbeats (periodic re-dispatch of the stored setpoint) are the one non-governed re-dispatch: off by default, stopped by line stop or driver change, and re-applying the stored value without re-admission—plant-facing state maintenance, not an agent path. Line start installs active context before recipe parameters dispatch individually; line stop clears it.

C. Write Acknowledgment as Three-Level Evidence

A driver-level success flag conflates two facts: the transport accepted the command, and the device now holds the value. AGENTWORKSHOP grades every governed write at two checkpoints. At *acceptance*, the driver classifies the write as *readback-verified* (its own post-command read returns the commanded value within the node's configured write tolerance), *transport-acknowledged* (the transport accepted the command, nothing device-side observed yet—a broker receipt or an MES business acknowledgment), or *unverified* (the transport failed). At *verification*, a post-write verifier re-reads the endpoint through the node's read path (three attempts, 2 s apart) and promotes transport-acknowledged parameters whose read returns the commanded value; a write that cannot be confirmed closes *unverified*, and one whose command never completed acceptance closes *failed*—batch receipts carry a per-parameter grade plus a three-bucket summary (verified/unverified/failed), where the summary counts verification outcomes and the transport-acknowledged terminal grade lives in the per-write journal of acceptance-receipt routes; the write journal stores the per-write class, and the operations log records worst-state severity. The falsifiable contrast: under a success-flag receipt rule—a transport acknowledgment closes the report—the publish-only receipt of Section V-G (broker puback, `ok=true`, no read channel) reports *success*; the graded rule reports it *unverified*. Two boundary cases fix what the grades mean: a MES write whose endpoint returns an explicit application-level acceptance but exposes no readback mapping terminates transport-acknowledged—an acceptance receipt, still short of device confirmation—whereas a broker receipt alone terminates unverified. Absence of evidence degrades the grade, never the report; the grade is fixed when the write closes.

A transport-acknowledged write may still not have reached the device; promotion through the verifier is what turns acceptance into confirmation, and the three-way check of Section IV-D reads the value back once more from the acquisition store. Recipe rollback restores a definition, not physical PLC values, and a compensating write is itself graded by the same rule—readback-verified where the endpoint supports readback, at best transport-acknowledged where only an acceptance receipt exists. The grade closes the command path only: readback-verified says the device holds the value, not that the process responded as intended—that distinction belongs to observation and verdict.

D. Observation, Verdict, and Compensation

An agent- or rollback-source write opens an optimization record; the verdict tool `dcw_judge` records `keep`, `rollback`, or `uncertain` plus a reason; the verdict, stored with the record it closes and the response window actually queried, performs no actuation and does not independently validate the evidence, so a scripted keep can coexist with a missed objective. Verdict integrity therefore rests on attribution, journaling, and operator review—a kept-but-marked record stays visible with its disagreement attached, and the operations log raises parameter severity under alarm-rationalization practice [31]—with the next guard queued: an independent adjudication identity, or a platform-side minimum re-check before a caller's `keep` may stand down the backstop (Section VI). Three-way setpoint consistency is checked by the evaluation harness rather than the platform: the harness compares the recipe value, the device readback, and a mirror sample drawn from the acquisition runtime's separate polling path and store, each leg against a stated tolerance; the sampled legs meet at the plant endpoint, so a fault fabricating the endpoint is their common failure mode, and disagreement is reported with the run. Verdicts are attributable—actor and timestamp travel with them.

Compensation is a new, separately admitted write that may fail or partially complete; it is graded by the Section IV-C rule, and a rollback record settles to rolled back once the compensating write is accepted. If the compensating write is itself refused by admission, the source record closes *as judged*, with the refusal appended to its verdict. These separations keep command status, record status, and process recovery independently inspectable.

The backstop is a record-integrity check for writes that no verdict has closed—not an independent protection layer: it evaluates only open records; a caller's `keep` closes and stands it down, while a `rollback` verdict stands it down only once

the compensating write executes—availability is therefore contingent on recorded verdicts and arriving observations. When collected samples of an open record's node leave the recipe's monitoring window (a DAQ-side envelope distinct from the admitting interval), the platform issues a system-attributed rollback verdict and restores the value preceding that write (one step; the admitted baseline is not restored), graded as above. The policy waits 120 s, polls every 30 s, and fires on three out-of-window samples—cumulatively, not necessarily consecutively; non-returning samples count no breach, so an acquisition outage leaves the record open and unalarmed. Recovery runs only while fewer than two retained rollback records stand on the node; past that, the backstop judges but defers actuation to the approval policy.

E. Human Approval Boundary and the Line Master Switch

Manual bindings insert the timed approval wait before an agent setpoint or compensation request; automatic bindings omit the wait but keep the checks. The pending request carries member, node, value, interval, expiry, the proposer's reasoning, and per-parameter bases; Fig. 3b shows the card's value row, reasoning line, and decision box—machine-checked fields as structured data, the free text display-only and non-authoritative. Because binding and admission are re-checked at execution, approval reserves neither interval nor device state; the approval dialog—itself an injection surface (Section VI)— cannot promote a proposal past the checks; the comment returns verbatim to the proposing member on refusal and travels with the receipt on approval—feedback, not authority. The wait fails closed by default (expiry 180 s): an expired request resolves as refusal and is recorded; duplicates are suppressed by holding one pending request per member–node pair; and in *hold* mode an unanswerable request persists without expiry until a human answers.

Human authority is anchored one level above individual requests: each line carries a control mode—the line master switch—whose fail-safe default is *manual*. Under manual mode, recipe-family operations (parameter update, dispatch, trial, and rollback) from approval-free bindings on that line are intercepted and presented for approval regardless of binding policy; direct node-tool writes remain gated by their binding mode and the admission interval, not by the line switch. Switching a line to automatic requires an explicit admin-only user-side confirmation; agent identities cannot touch the switch. The residual agent-side path is delegation: a lead agent can grant held nodes to a worker through an attributed lead-only tool; the grant's execution mode is chosen at grant time rather than gated by the user-side confirmation, is attributed to the granting agent, cannot exceed its own node face, and is individually revocable (Section VI).

The decision endpoint authenticates a user but does not enforce a line-scoped approver role—the gap Section VI carries—and reads, verdict recording, and some configuration changes do not use physical-write approval—workflow mediation, not segregation of duties; a recipe switch between approval and execution re-admits the value against the new window. The strategy engine is replaceable rather than trusted as a control primitive: scenario code declares and configures but never executes plant I/O, and the platform never infers a process objective.

V. Experimental Evaluation

The evaluation asks whether one governed supervisory path can be exercised, inspected, and repeated across heterogeneous industrial jobs, not whether one optimizer beats another. Five questions organize the evidence: protocol closure and admission-gate attribution (Q1), the agent-team workflow (Q2), a deterministic controller loop (Q3), four concurrent missions (Q4), and deployment-scale regression with write-evidence verification (Q5). Q2–Q4 occupy the strategy slot (scripted policies, with two live engine turns in Q2); Q5 stresses the hardened platform under load. Together the five exercise each production-facing plane of Section III.

A. Testbed, Evidence Snapshot, and Scope

The evaluation draws on two evidence tiers of one codebase. The *archived snapshot* for Q1–Q4 is integrated run `20260921184204-10lc` (seed 42, commit `e692df1`, harness `9f2756dce4a4cbb3`, completed September 21, 2026, 19:08 UTC), archived with raw reports, plus ablation companion `20260921181234-e1lite`. Its standing preset is a cast-film reference line tuned through the die-head heater setpoint (objective J weights thickness conformance, defects, energy, and throughput). Each mission contributes one seed-42 trajectory: Q1, Q2, and Q4 are single-trace case studies, checkable claim by claim in the artifacts; the deterministic loop adds seeds 43–44, and repeat evidence comes from Q5's deployment-tier campaigns—a different tier, not a re-run.

The *deployment-scale tier* for Q5 is a live v0.7.57 instance carrying 34 provisioned lines, 111–113 agent channels, 67 online DAQ nodes, 77 write nodes, 52 recipes, and 385 member–node bindings—simulator- and bench-backed objects, not physical plants; *multi-line* means many provisioned lines and channels on one host. The tier's benchmark line—a different plant from that cast-film reference—is a physics-based injection-molding cell (16 DAQ nodes, 6 control nodes over five protocol families plus MES, four exercised for governed writes); all Q5 numbers come from committed per-run reports, ledgers, and timelines (October 7: three baseline-suite runs; October 8: the hardened-pipeline run plus two full-suite runs; October 9: three consecutive regression runs; October 12: four control-pair replays); the campaign ran under live load. One timeline fact matters: acknowledgment grading and the master switch came with the October hardening pass, *after* the Q1–Q4 snapshot—Q1–Q4 therefore evidence the governed path without them, and the grading and master-switch surfaces are evidenced by Q5 (the seven-leg suite and every campaign dispatch); the ablation isolates the window interlock, untouched by that pass. The harness sets the benchmark line's master switch to *automatic* by user-side confirmation—the automated path is the object under test—and verifies manual-mode interception separately (legs 1 and 5).

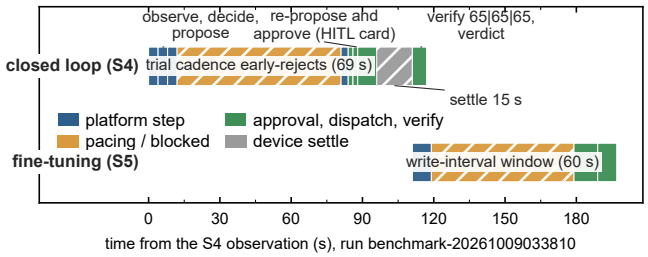


Fig. 4. One governed iteration on the October 9 line: a corrective decision (0.276 g error) packaged by the proposer as a sensitivity excitation; the trial cadence early-rejects 69 s before a human is involved; structured approval releases the re-proposal, dispatch returns readback-verified, and the three-way check reads 65—65—65. Lower lane: fine-tuning under the same 60 s window, rolled back and restored.

B. Test Procedure and Campaign Overview

The Q1–Q4 procedure is a 20-phase integrated suite—an author-defined regression instrument, not an external certification—one claim per phase, from preset loading through protocol interception and the record, recipe, approval, and parameter-layer drills (Sections V-C and V-F).

The Q5 procedure is the one-command benchmark pipeline: eight stages (S0 pre-check; S1 API-surface matrix across the functional surface of Sections III–IV; S2 multi-modal ingestion with MES dual-mode; S3 governance negative paths; S4 closed-loop optimization; S5 fine-tuning; S6 diagnostics; S7 report) under a harness capturing every check and a timestamped timeline per run; the October 9 campaign re-runs it three times. No prior system in Table I owns a governed write path to plant endpoints, so the primary comparator is internal (Section VI records the next evidence tier). Two controls are archived. The admission control relaxes the mechanisms: the six window-edge probes that full governance rejects (HTTP 400) return 200 *executed* under the relaxed arm (Table II)—a success-flag design reports accepted writes. The evidence control holds the receipts and varies the reporting rule (Section IV-C); its success-flag outcome is definitional—the pre-hardening harness reported a broker puback as success. The closed-loop harness also parameterizes the write injection (admin REST beside the agent's governed `dcw_control`); a like-for-like pair—the same writes to both routes—replayed in four archived post-campaign runs per route (October 12) was rejected in full, 24/24 per arm with and without production pacing—gate evidence, not route quality; a quality comparison awaits a dedicated instance. Fig. 4 summarizes the closed-loop instrument.

C. Q1: Protocol Closure and Admission-Gate Attribution

On the five protocol lines the suite stores 55 DAQ samples through real drivers and dispatches governed writes over four protocol families—about 6 writes each (19.3–47.8 ms p50, 20.1–248.9 ms p95; per-protocol values in the archived metrics); the RTU-over-TCP satellite is acquisition-only—readback confirmation where supported—with one admission, record, and verdict path for every request. Each writable line rejects 6/6 out-of-window probes (24/24) while accepting its legal write. Two drills separate truth paths one success

TABLE II
ADMISSION ABLATION ON COMPANION RUN `20260921181234-e1lite`: SIX PROBE CLASSES PER ARM, THREE DETERMINISTIC REPETITIONS. THE FOUR HARD-RANGE CLASSES ARE REJECTED IN EVERY ARM; ONLY THE TWO WINDOW-EDGE CLASSES FLIP.

Arm	Soft win.	Readback	Intercept	Executed
full	✓	✓	18/18	0
no-readback	✓	—	18/18	0
no-interlock	—	✓	12/18	6 jour.
ungated	—	—	12/18	6 jour.

flag would collapse: a frozen-measurement drill (write and readback consistent, measurement out of window) ends with a system-judged backstop rollback 130.2 s later, and a record-rollback drill restores the prior value by a separately admitted, readback-confirmed write. A second preset with different signals and dynamics is commissioned by configuration alone; across both presets 33 probes are rejected on seven writable lines (24+9)—the same admission code discriminating on unseen signal maps. The parameter layer makes the interval geometry of Eq. (1) observable (Fig. 3a): four declared layers narrow to one admitted interval, each rejection attributable to its excluding layer; a manual binding suspends a request visibly and unblocks it on approval (latency 461 ms, harness-adjudicated).

The ablation replays the same probes on companion run `20260921181234-e1lite` and a standalone fixture (six probe classes per arm, three deterministic repetitions). Table II reports the four arms: full governance and the no-readback arm reject 18/18; without the window interlock (alone or with the readback check) only 12/18 reject and the six window-class probes execute and are journaled—prevention is the interlock's job; a relaxed configuration degrades from prevention to detection, with the readback check retained to make written-but-unconfirmed commands visible. The hard range is structurally unbypassable—all four governed routes re-admit it at one journaling entry point, per request (Section IV-B)—and the soft window is its removable part.

D. Q2: The Agent-Team Optimization Workflow and Its Results

The workflow ran end to end on the cast-film line and then a larger BOPET line (Section III-B); a scripted policy issued the writes (two live engine turns follow). On the single-node mission one write sufficed ($200 \rightarrow 204.704\,^{\circ}\mathrm{C}$ against target $204.704 \pm 0.75\,^{\circ}\mathrm{C}$, settled at $204.70\,^{\circ}\mathrm{C}$, journal-attributed, task `COMPLETED`). The BOPET mission is the multi-node case: nine devices and 49 signals (30 writable, 19 readable), all ensure-pass created. Three governed writes on three distinct nodes—casting speed $32 \rightarrow 33.8$ m/min, fast-roll speed $118 \rightarrow 120.3$ m/min, and rail width $3000 \rightarrow 3035$ mm— bring thickness from 28.10 to 25.66 μm against $25.0 \pm 0.7\,\mu$m—$0.04\,\mu$m inside the window; the same run's prose summary records it *not* attained while its machine-readable verdict records attained, so we report both; the verdict matches the archived predicate, whose terminal observation sits inside the window on re-check—the prose narration erred. Each write opened a record, received `keep`, and left an agent-attributed

journal entry; the parent task reaching `COMPLETED`. Finally, two live agent-engine turns (omp adapter, glm-5.3-flash) drove the same slot on a scratch line—a bounded write (156.3 s), a goal-driven descent settling at 51.7 μm against a harness-supplied target ($52.0 \pm 0.8\,\mu$m, in the task text, not the scenario predicate), and a recipe save with attributed authorship (637.3 s). They show that an engine can occupy the slot, not agent quality—the run's pass criterion is a completion marker in the transcript.

E. Q3: The Deterministic Closed Loop

Seeds 42–44 each issue two governed writes with no rejection, raising the cast-film objective J from $J_0 \approx 67.2$ toward its grid optimum ($J/J^* = 0.965$–0.972 against $J^* = 89.894$; the grid enumerates the admissible zone-temperature, screw-speed, and line-speed setpoints, so J^* lower-bounds the true optimum and the reported ratio upper-bounds J/J_{true}). What the benchmark establishes is narrower: the governed path passes a conventional controller's writes through unmodified—admissible, journaled, read back; the claim is transparency of evidence handling, not a performance delta.

F. Q4: Four Process Missions

The campaign runs four blueprint-provisioned lines concurrently; in this run (seed 42) all four attain their archived predicates—evaluated from subsequent observations, not task state. Fig. 5 archives each mission's optimization trajectory. Injection molding is the minimal case (two writes: hold pressure 45→58.2 bar, hold time 6→10.64 s), BOPET the multi-node case of Section V-D; both close with task state and predicate agreeing. Wastewater treatment (anaerobic–anoxic–oxic) is the write-heaviest mission (15 governed writes across distinct nodes—the Section IV-B spacing is per node): dissolved oxygen rises from 0.21 into 3.4 ± 0.6 mg/L (attainment window; terminal observation 3.50 mg/L, Fig. 5b) under five effluent guards: four guard margins finish wider than at baseline, and the fifth, pH, drifts 6.59→6.38. Its most informative events are blower steps that were admissible and journaled yet lost compliance margin; the verdict stage reverted them—a bad proposal is rejected in the evidence, not at the admission gate. Annealing starts far outside the hardness window (134.52 HV at 140 m/min); zone-3 temperature, one zone-2 trim, and two line-speed steps hold 95 ± 6 HV while raising line speed by 17.1%, and a further attempt at 176 m/min is reverted for leaving the margin—on record. Mission clocks (26.48 s, 169.27 s, 102.76 s; BOPET's unarchived) are simulated mission durations under accelerated dynamics; across the four missions in this run, platform-side behavior is identical while the write economy—governed writes per scripted policy—varies by more than a factor of seven (2, 15, 5, and 3).

G. Q5: Deployment-Scale Regression and Write-Evidence Campaign

The final lens stresses the platform: the hardening pass made the evidence grading, the master switch, and the approval-anchored dispatch interval of Section IV the explicit

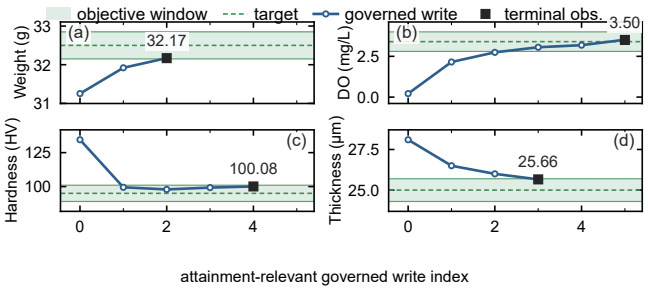


Fig. 5. Archived optimization trajectories of the four concurrent missions (seed 42): observed controlled quantity after each attainment-relevant governed write, against the archived objective window; the index counts those writes ((b) plots 6 of its 15). (a) Injection molding; (b) wastewater treatment (A2O); (c) annealing; (d) BOPET film. Dissolved oxygen; observations under scripted policies.

TABLE III
CAMPAIGN PROGRESSION ACROSS THE OCTOBER PASSES; EVERY EXECUTION COMPLETES WITH ZERO FAILED CHECKS.

Executions	Test surface	Per run	Total
October 7 (3 runs)	baseline S0–S7 suite	79	237
October 8 (3 runs)	+ grading, master-switch checks	90–92	273
October 9 (3 runs)	same surface under live load	90–92	272
nine executions	zero failed checks	—	782

test surface of the closed-loop, fine-tuning, and write-evidence scenarios. *Full-suite repeatability.* Nine pipeline executions complete with zero failed checks (Table III); October 7–8 runs wait out the remaining 218 s inside the per-line trial cadence—pacing stays in force under automation.

Closed-loop scenario under governance. Each run walks observation, decision (dead-band 0.15 g, slope 0.052 g/bar), a five-element proposal, HITL approval, batch dispatch, and three-way consistency (recipe, PLC readback, mirror). The loop runs in two regimes: October 7–8 descends $66 \to 63$ bar one governed step per run; October 9 answers a downward drift in the mirrored weight ($32.49/32.33/32.22$ g) with a bounded zigzag (64↛63 bar) and two corrective steps (63↛65 bar) answering residual error ($0.169/0.276$ g)—the zigzag is deliberate excitation, not a correction. Every step verifies equal values across the three sources at maximum consistency error 0.0 bar (0.05 bar device, 0.51 bar bucketed-mirror tolerances—all legs equal at the endpoint's recording resolution). The pacing window acts before the human: in two October 9 runs the proposal is early-rejected by the trial cadence and re-proposed 69 s later; only then is the HITL card raised and structured approval releases the readback-verified dispatch (Fig. 4). Approval-to-dispatch takes 6–9 ms; observation-to-verification spans 25.1 s without a pending cadence window and 96.1 s on October 9, 245.20/245.18 s with one (October 7)—the loop's time is governance, not platform overhead. The fine-tuning scenario rides the same windows (Fig. 4, lower lane); the diagnostics scenario exports a 30-min merged wide table (663–869 rows)—a data-quality observation, not a process excursion.

Latency and mission figures are single-run point estimates (about 6 writes per line, one trajectory each); the Q1 approval latency is the harness's mediation span, not a human-response measurement, and Q5's approval-to-dispatch span is server-

side only. Only Q3 repeats across seeds, Q5's repetition is across consecutive deployment-tier campaigns, and the Q1 ablation repeats deterministically (Table II); the campaign re-exercises the governance surfaces (S1, S3) nine times with zero failures but does not re-run the Q1–Q4 missions, which remain single-trace case studies, as are the fault-suite grade outcomes (n=1–2 injected cases).

Write-evidence legs. A dedicated seven-leg suite verifies Sections IV-C and IV-E directly in legs (1)–(5): (1) the master switch refuses an unconfirmed manual-to-automatic transition; (2) a mock-acknowledged (transport-acknowledged) dispatch is promoted to readback-verified by post-write re-read; (3) a broker cut marks the affected parameter *failed* at the batch summary (operations level error) while its healthy sibling verifies; (4) a publish-only endpoint accepts the write (broker puback, `ok=true`) yet stays *unverified* through three verification attempts—the receipt the Section IV-C rule is stated against; (5) on a manual-mode line an approval-free binding's dispatch is intercepted into the queue, approval releasing all six parameters device-verified; (6)–(7) API regression and documentation-consistency suites close the campaign.

Observed grade distribution. In fault-free operation, every batch dispatch of the six hardened runs (October 8–9) closed *readback-verified*—structured approval released it and the three-way check read equal values at 0.0 bar maximum error (Fig. 4). The non-verified grades appear only under the fault suite's injected conditions, which is the distribution a healthy platform should produce: 1 *failed* (broker unreachable; healthy sibling verified), 1 *unverified* (publish-only endpoint, three attempts), and 2 parameters promoted from transport-acknowledged receipts to readback-verified by post-write re-read. The remaining terminal grade, transport-acknowledged, occurs on MES-style endpoints that return an application-level acceptance without a readback mapping; the platform's MES route tests cover it, but the archived campaign instantiates MES in read mode only—a disclosed gap in the campaign's write-evidence coverage.

Across the five questions, Section IV's mechanisms behave as specified: interception with attribution, honest grading, pacing under automation, mediated approval, and on-record recovery.

VI. Discussion and Limitations

A. Validity and Reproducibility

Run identifier, seed, commit, harness fingerprint, and raw artifacts are fixed, so counts are re-derivable rather than trusted; Q5 adds per-run ledgers and timelines. Five qualifications bound what they support: simulation-tier evidence (deterministic per seed; no physical-device campaign, operator study, or external comparison reported); internal-oracle optima and predicates under an author-implemented, author-adjudicated harness and simulator set; single-host latencies and near-contemporary runs; implemented-but-unexercised capabilities and a single live provider/model; and harness-adjudicated approval (human decision latency and quality out of scope).

B. Safety Positioning and Human Oversight

The threat model is a misbehaving or manipulated agent inside an otherwise trusted platform and plant; approval dialog, admission re-checks, and journaling are the containment surfaces; no injection campaign or approval-dialog operator study is reported; stolen credentials and compromised endpoints are the domain of IEC 62443 [23], and high-risk-AI regulatory duties—record keeping and human oversight—belong to instruments such as Regulation (EU) 2024/1689 [32]; the attributed journal and approval gates serve its logging and oversight articles at the evidence layer, certification out of scope. Fast regulation, hardware interlocks, and safety functions remain in the PLC; prompts and retrieved documents are untrusted text, not authority [33], [34]—the proposer is confined to per-binding scope and journaled, attributable actions: a bounded blast radius, not a protection guarantee. Admission limits, acknowledgment grading, and the backstop run with or without human approval—enforcement never queues behind human attention (the deferred-mode exception hands actuation back to the approval policy, Section IV-D). The pacing windows bound approval demand per line, and batch release (Section V-G, leg 5) is the demonstrated fatigue mitigation; approval fatigue and automation bias remain human-factors risks [30], [35]. A lead agent's delegation-mode choice (Section IV-E) is the one agent-reachable path around per-request approval; capping the granted mode at the granter's own binding mode, or requiring the line-switch confirmation for it, is queued as the guard, alongside the verdict-side guard of Section IV-D (an independent adjudication identity, or a platform-side minimum re-check before `keep`).

C. Deployment Boundaries

Against the nearest systems of Section II, the differentiator is not the LLM—a replaceable strategy slot—but the production-scoped authority and evidence around its writes: no nearest system reports that combination on one path. The layer installs beside the plant's protocol endpoints, onboards declaratively (Section III-B), and leaves PLC protection and DCS interlocks untouched. Execution and recovery are non-atomic: bookkeeping may fail after actuation, the pacing windows time writes rather than certify stability, declared per-write step limits are advisory metadata surfaced to agents rather than enforced bounds, and multi-node compensation may partially succeed. Readback agreement and three-way consistency need not imply process recovery; hold heartbeats re-apply the stored setpoint without re-admission, trailing a narrowed window. Oscillation, storage failure, clock error, concurrency beyond the four-line campaign, and split-brain supervisor deployments with divergent stores remain unevaluated; the journal is append-only but not tamper-evident—no hash chain or trusted timestamping—so forensic integrity rests on host security, and clock skew propagates into timeline evidence. A supervisor outage halts dispatch rather than permitting writes through; plant staff retain their conventional DCS/SCADA stations outside AgentWorkShop—the governed path mediates agent-originated and platform-client writes, not existing operator tooling: an integration boundary, not a takeover. The next evi-

dence tiers are a hardware-in-the-loop leg against a real PLC or OPC-UA endpoint and community benchmarks such as Tennessee Eastman [36] or Secure Water Treatment [37]. Production integration additionally requires change-controlled bound changes with process-evidence validation, approval/open-record aging and backstop-deferred states surfaced on the plant alarm system, the bench-mode switch refused at production boot, and approval as workflow mediation—attribution and timestamp, not an electronic signature. In production, bindings govern sanctioned tools, not arbitrary privileged-engine I/O—least privilege, endpoint security, and IT/OT separation remain necessary [23]; AGENTWORKSHOP is neither a safety instrumented function nor a plant-protection substitute [38]; line-scoped approver authorization is still missing, so production needs segregation of duties and alarm-path failure tests [31].

VII. CONCLUSION

AGENTWORKSHOP contributes a governed supervisory execution layer rather than a new process optimizer: one governed path from agent-team identities through semantic nodes and production context to governed writes, graded acknowledgments, verdicts, and compensation—admission, approval, and recovery left as separately inspectable, actor-attributed evidence on every agent-originated write. Physical-plant deployment, operator studies, and external-framework comparison remain open; until then, optimization benefit, autonomous control, and safety assurance stay unsupported claims.

Data availability and disclosure. Artifacts are archived per run in the project's benchmark result tree; the platform and its deterministic simulators are publicly available (Section III). The live engine-turn transcripts (Section V-D) were excluded by the log-retention rule and survive only as recorded outcomes; the write-evidence suite is rerunnable from its script. AI assistants supported editing under author direction; the authors remain responsible for all claims.

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